

## GIMPO INTL, Korea, Republic of

WMO: 471100

Lat: 37.558N

Lon: 126.791E

Elev: 18

StdP: 101.11

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.2      | -11.2 | -23.2                         | 0.5 | -8.8 | -20.2 | 0.6 | -7.0 | 8.9                   | -4.6 | 7.9 | -3.1 | 2.4                   | 330  | 0.450 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 7.6                    | 32.9            | 25.2 | 31.2 | 24.4 | 30.0 | 23.5 | 26.7                | 30.0 | 26.0 | 29.2 | 25.3 | 28.3 | 3.6                  | 290  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.0                            | 21.3 | 28.3 | 25.1 | 20.2 | 27.6 | 24.2 | 19.1 | 27.0 | 83.8          | 30.6 | 80.7 | 29.0 | 77.7 | 28.6 | 29.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 8.2           | 7.1  | 6.2 | DB                         | -16.0 | 34.7               | 2.8 | 1.3                                                | -18.0 | 35.6       | -19.6 | 36.4       | -21.2 | 37.1       | -23.2 | 38.0 | (4) |
| (5)               |      |     | WB                         | -16.8 | 27.4               | 2.7 | 1.0                                                | -18.7 | 28.1       | -20.2 | 28.7       | -21.7 | 29.3       | -23.7 | 30.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.1        | -3.4  | -0.6  | 5.0   | 11.6  | 17.5  | 22.3  | 25.4  | 26.2  | 21.2  | 14.2  | 6.4   | -1.0  | (6)  |  |
|                                                                        | DBStd       | 10.77       | 4.24  | 3.99  | 3.55  | 3.51  | 2.54  | 2.06  | 2.12  | 2.47  | 2.68  | 3.63  | 4.37  | 4.27  | (7)  |  |
|                                                                        | HDD10.0     | 1369        | 417   | 298   | 159   | 23    | 0     | 0     | 0     | 0     | 0     | 9     | 123   | 341   | (8)  |  |
|                                                                        | HDD18.3     | 2966        | 675   | 532   | 413   | 203   | 46    | 1     | 0     | 0     | 7     | 132   | 358   | 599   | (9)  |  |
|                                                                        | CDD10.0     | 2146        | 0     | 0     | 4     | 71    | 233   | 368   | 477   | 502   | 334   | 141   | 15    | 0     | (10) |  |
|                                                                        | CDD18.3     | 701         | 0     | 0     | 0     | 1     | 21    | 119   | 219   | 244   | 92    | 6     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 5522        | 0     | 0     | 0     | 19    | 176   | 743   | 1746  | 2257  | 550   | 31    | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 1647        | 0     | 0     | 0     | 2     | 23    | 173   | 533   | 813   | 102   | 2     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.7         | 2.5   | 2.7   | 3.1   | 3.1   | 3.0   | 2.7   | 2.9   | 2.7   | 2.4   | 2.4   | 2.6   | 2.6   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1441        | 18    | 29    | 41    | 92    | 102   | 150   | 426   | 306   | 160   | 49    | 45    | 23    | (15) |  |
|                                                                        | PrecMax     | 2309        | 59    | 64    | 109   | 176   | 281   | 499   | 1064  | 936   | 572   | 217   | 106   | 65    | (16) |  |
|                                                                        | PrecMin     | 804         | 2     | 1     | 6     | 31    | 13    | 46    | 156   | 40    | 16    | 0     | 8     | 3     | (17) |  |
|                                                                        | PrecStd     | 354         | 14    | 20    | 26    | 35    | 56    | 99    | 203   | 221   | 142   | 44    | 25    | 16    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 9.2   | 13.9  | 18.9  | 25.9  | 29.1  | 31.8  | 34.1  | 35.1  | 30.2  | 26.1  | 20.0  | 12.0  | (19) |  |
|                                                                        |             | MCWB        | 6.0   | 9.2   | 10.9  | 14.5  | 17.1  | 20.1  | 25.5  | 25.6  | 22.4  | 17.7  | 14.3  | 7.8   | (20) |  |
|                                                                        | 2%          | DB          | 7.0   | 10.1  | 16.0  | 22.9  | 27.0  | 30.0  | 32.1  | 33.2  | 29.1  | 24.0  | 17.2  | 9.8   | (21) |  |
|                                                                        |             | MCWB        | 3.3   | 5.4   | 8.9   | 14.1  | 16.6  | 20.5  | 25.1  | 25.5  | 21.5  | 16.8  | 12.5  | 6.3   | (22) |  |
|                                                                        | 5%          | DB          | 5.0   | 8.0   | 13.9  | 20.9  | 25.2  | 28.8  | 30.9  | 32.1  | 27.9  | 22.2  | 15.9  | 7.9   | (23) |  |
|                                                                        |             | MCWB        | 1.8   | 4.1   | 7.8   | 12.3  | 16.2  | 20.0  | 24.6  | 25.0  | 20.6  | 15.8  | 11.6  | 5.0   | (24) |  |
|                                                                        | 10%         | DB          | 3.2   | 6.1   | 11.9  | 18.8  | 23.9  | 27.2  | 29.8  | 30.9  | 26.2  | 20.9  | 14.0  | 6.0   | (25) |  |
|                                                                        |             | MCWB        | 0.6   | 2.5   | 6.5   | 11.6  | 15.7  | 19.7  | 23.9  | 24.8  | 19.7  | 15.0  | 10.1  | 3.3   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 6.2   | 10.0  | 12.8  | 18.0  | 20.5  | 24.2  | 27.4  | 27.8  | 25.8  | 20.4  | 16.0  | 9.0   | (27) |  |
|                                                                        |             | MCDB        | 8.7   | 11.5  | 16.5  | 21.2  | 24.2  | 27.1  | 30.3  | 31.1  | 27.9  | 22.8  | 18.3  | 10.8  | (28) |  |
|                                                                        | 2%          | WB          | 3.9   | 6.1   | 10.4  | 15.5  | 19.1  | 22.8  | 26.5  | 26.9  | 23.9  | 18.6  | 14.0  | 6.7   | (29) |  |
|                                                                        |             | MCDB        | 5.8   | 8.9   | 13.9  | 19.7  | 23.2  | 25.9  | 29.6  | 30.5  | 26.6  | 21.8  | 16.4  | 8.8   | (30) |  |
|                                                                        | 5%          | WB          | 2.4   | 4.3   | 8.6   | 13.9  | 18.0  | 21.9  | 25.8  | 26.3  | 22.6  | 17.4  | 12.1  | 5.0   | (31) |  |
|                                                                        |             | MCDB        | 4.4   | 7.2   | 12.2  | 18.9  | 22.5  | 25.6  | 28.8  | 29.8  | 25.3  | 20.5  | 15.1  | 7.2   | (32) |  |
|                                                                        | 10%         | WB          | 0.9   | 2.9   | 7.0   | 12.6  | 17.0  | 21.2  | 25.1  | 25.7  | 21.5  | 16.0  | 10.5  | 3.5   | (33) |  |
|                                                                        |             | MCDB        | 2.9   | 5.5   | 10.6  | 17.0  | 21.3  | 24.8  | 27.9  | 29.1  | 24.6  | 19.3  | 13.3  | 5.7   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.8   | 10.5  | 10.9  | 11.5  | 10.9  | 9.3   | 6.9   | 7.6   | 9.6   | 11.4  | 10.2  | 9.4   | (35) |  |
|                                                                        |             | MCDBR       | 10.7  | 12.6  | 13.6  | 14.5  | 13.5  | 11.7  | 8.9   | 9.1   | 10.5  | 12.2  | 11.8  | 11.1  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 7.8   | 8.4   | 7.7   | 6.8   | 5.7   | 4.6   | 3.7   | 3.5   | 4.4   | 6.2   | 7.8   | 8.3   | (37) |  |
|                                                                        |             | MCWBR       | 9.6   | 11.0  | 11.5  | 11.7  | 10.5  | 8.9   | 7.1   | 7.7   | 8.4   | 10.5  | 10.4  | 10.4  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.437 | 0.491 | 0.571 | 0.618 | 0.631 | 0.704 | 0.672 | 0.584 | 0.514 | 0.506 | 0.486 | 0.436 | (40) |  |
|                                                                        |             | taud        | 1.947 | 1.853 | 1.717 | 1.666 | 1.690 | 1.613 | 1.726 | 1.931 | 2.046 | 1.940 | 1.931 | 1.972 | (41) |  |
|                                                                        | Ebn at Noon |             | 695   | 709   | 693   | 693   | 694   | 644   | 661   | 712   | 739   | 688   | 639   | 661   | (42) |  |
|                                                                        |             | Edh at Noon | 139   | 173   | 218   | 242   | 240   | 260   | 230   | 184   | 155   | 157   | 141   | 127   | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 2.51  | 3.28  | 4.30  | 5.00  | 5.62  | 5.26  | 4.11  | 4.40  | 4.17  | 3.60  | 2.44  | 2.19  | (44) |  |
|                                                                        | RadStd      |             | 0.16  | 0.26  | 0.42  | 0.34  | 0.53  | 0.56  | 0.63  | 0.62  | 0.45  | 0.25  | 0.24  | 0.15  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |  |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|--|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |  |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |      |  |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +133    | +77     | (46) |  |
| (47) Regional (47 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +92     | +50     | (47) |  |

Nomenclature: See separate page